

Nonexistence of a Four-Layer Balanced Hamilton Starter in $\text{Cay}(\mathbb{Z}_{12}, \{1, 2, 3\})$

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Abstract

For the directed circulant $C_4(3) = \text{Cay}(\mathbb{Z}_{12}, \{1, 2, 3\})$, we prove that no four-layer balanced directed Hamilton cycle exists. Balance turns the problem into a finite choice of one arc from each of twelve translation orbits. An exact exhaustive reduction leaves only 15 selections having indegree and outdegree one at every vertex; twelve have cycle type $6 + 6$ and three have cycle type $4 + 8$. Thus none is Hamiltonian. Consequently, the more restrictive request that the three developed cycles contain $0 \rightarrow 2$, $2 \rightarrow 4$, and $4 \rightarrow 6$ in distinct cycles also has a negative answer. The result concerns only $q = 4, r = 3$ and makes no assertion for other values $r \equiv 3 \pmod{4}$.

1 Statement and context

All vertex arithmetic below is modulo 12. The directed circulant $C_4(3)$ has vertex set $\mathbb{Z}_{12}$ and arcs $x \rightarrow x + s$ for $s \in \{1, 2, 3\}$. Its layer $j \in \mathbb{Z}_4$ consists of the vertices congruent to j modulo 4. Following Jiang and Yang [1], a spanning subdigraph is *four-layer balanced* when, for every $(j, s) \in \mathbb{Z}_4 \times \{1, 2, 3\}$, it contains exactly one step- s arc whose tail belongs to layer j . A balanced Hamilton cycle is a four-layer balanced Hamilton starter. Its translates by 0, 4, 8 would form a Hamilton decomposition [1, Proposition 2.2 and Lemma 2.3].

The closest inspected construction gives four-layer starters for $r \equiv 1 \pmod{4}$, $r \geq 9$ [1, Theorem 4.7]. The same paper explains that its particular ABAB construction fails when $r \equiv 3 \pmod{4}$, while explicitly distinguishing that failure from a general existence obstruction [1, Section 9]. The present note addresses only the smallest instance $r = 3$: its contribution is the complete finite obstruction below. It neither claims a result for any other r nor makes a literature-priority claim.

Theorem 1. *There is no four-layer balanced directed Hamilton cycle in $\text{Cay}(\mathbb{Z}_{12}, \{1, 2, 3\})$.*

Corollary 2. *There is no four-layer balanced Hamilton starter whose three translates admit one selected arc each such that the selected arcs form $0 \rightarrow 2 \rightarrow 4 \rightarrow 6$.*

2 Exact finite reduction

For $j \in \mathbb{Z}_4$ and $s \in \{1, 2, 3\}$, let

$$O_{j,s} = \{(j + 4t) \rightarrow (j + 4t + s) : t \in \{0, 1, 2\}\}.$$

These are precisely the twelve arc orbits under translation by 4.

Lemma 3. *Balanced arc sets are in bijection with the 3^{12} arrays $(t_{j,s}) \in \{0, 1, 2\}^{\mathbb{Z}_4 \times \{1,2,3\}}$, where $t_{j,s}$ selects the arc*

$$j + 4t_{j,s} \longrightarrow j + 4t_{j,s} + s$$

from $O_{j,s}$.

Proof. Balance requires exactly one representative from each $O_{j,s}$, and every orbit has exactly the three displayed possible tails. The choices are independent and every balanced arc set determines them uniquely. $\square$

Lemma 4. *Exactly $(3!)^4 = 1296$ balanced arc sets have outdegree one at every vertex. They are parameterized by four permutations $\sigma_j : \{0, 1, 2\} \rightarrow \{1, 2, 3\}$ via*

$$f(j + 4t) = j + 4t + \sigma_j(t) \pmod{12}. \quad (1)$$

Such a selection also has indegree one everywhere if and only if the twelve values of f are distinct. In that case it is Hamiltonian if and only if f consists of one cycle of length 12.

Proof. For a fixed j , the three selected arcs have tails $j + 4t_{j,1}$, $j + 4t_{j,2}$, and $j + 4t_{j,3}$. They give the three vertices of layer j outdegree one exactly when the three ternary values are distinct. Equivalently, assigning to each tail index t its leaving step defines a permutation σ_j . Layers have disjoint vertex sets, so the four choices are independent. This gives $(3!)^4$ selections and formula (1). The remaining assertions are the standard image and cycle criteria for a map on a twelve-element set. $\square$

We now exhaust the 6^4 quadruples of permutations from Lemma 4. For each quadruple, compute the twelve successors in (1), retain it precisely when those successors are distinct, and decompose the resulting permutation by starting at the least unseen vertex. This uses only integer addition, reduction modulo 12, set cardinality, and permutation tracing.

Table 1: Exact counts in the exhaustive reduction.

Predicate	Count
Balanced selections	$3^{12} = 531441$
Outdegree one at every vertex	$6^4 = 1296$
Indegree and outdegree one at every vertex	15
Cycle type 6 + 6	12
Cycle type 4 + 8	3
One directed cycle of length 12	0

The complete list of the fifteen degree survivors appears in Appendix A. In particular, the table is an inspectable certificate that every survivor has two components. A second implementation also traversed the unreduced 3^{12} domain directly, in a different orbit order, and returned the same fifteen successor maps and the same cycle-type histogram. Appendix B gives the compact reduced verifier.

Proof of Theorem 1. By Lemma 3, every balanced arc set occurs in the finite domain. A directed Hamilton cycle must have outdegree and indegree one at every vertex, so Lemma 4 places it among the fifteen survivors in Appendix A. Twelve survivors are unions of two directed cycles of lengths 6 and 6, while the other three are unions of cycles of lengths 4 and 8. None is a directed 12-cycle. Hence no balanced Hamilton cycle exists. $\square$

Proof of Corollary 2. The requested developed cycles presuppose a four-layer balanced Hamilton starter. Theorem 1 shows that this base object does not exist, so the additional requirement on the three step-two path arcs cannot be satisfied. $\square$

Computational and AI assistance

Generative AI tools assisted with finite-search design, implementation of independent enumerators, checking of the written argument, and manuscript preparation. The decisive certificate is an exhaustive exact-integer calculation, not probabilistic model output: one program traverses all 3^{12} balanced selections, a separate program traverses them by an independent base-three counter and orbit order, and the reduced verifier below traverses all 6^4 necessary outdegree-one cases. No random sampling or floating-point arithmetic is used. The recorded calculations were run with CPython 3.12.14.

A All degree survivors

Each row gives the complete canonical directed-cycle decomposition of one selection having indegree and outdegree one at every vertex. Within a cycle, successive entries (and the last back to the first) are its directed arcs.

	First cycle	Second cycle	Type
1	(013467910)	(25811)	4 + 8
2	(0146910)	(2357811)	6 + 6
3	(013569)	(24781011)	6 + 6
4	(013479)	(25681011)	6 + 6
5	(0147810)	(2356911)	6 + 6
6	(0245811)	(1367910)	6 + 6
7	(0236811)	(1457910)	6 + 6
8	(0236910)	(1457811)	6 + 6
9	(0369)	(1245781011)	4 + 8
10	(0367911)	(1245810)	6 + 6
11	(025689)	(13471011)	6 + 6
12	(023568911)	(14710)	4 + 8
13	(034689)	(12571011)	6 + 6
14	(0358911)	(1246710)	6 + 6
15	(0346710)	(1258911)	6 + 6

B Compact reduced verifier

The following Python program directly implements Lemma 4. Its assertions are the last four rows of Table 1.

```

from collections import Counter
from itertools import permutations, product

orders = tuple(permutations((1, 2, 3)))
survivors = []
for by_layer in product(orders, repeat=4):
    f = [-1] * 12

```

```

for j, order in enumerate(by_layer):
    for t, step in enumerate(order):
        x = j + 4*t
        f[x] = (x + step) % 12
if len(set(f)) != 12:
    continue
unseen, cycles = set(range(12)), []
while unseen:
    start = min(unseen)
    x, cycle = start, []
    while x in unseen:
        unseen.remove(x)
        cycle.append(x)
        x = f[x]
    assert x == start
    cycles.append(tuple(cycle))
survivors.append(tuple(cycles))

types = Counter(tuple(sorted(map(len, cycles)))
                  for cycles in survivors)
assert len(orders)**4 == 1296
assert len(survivors) == 15
assert types == Counter({(6, 6): 12, (4, 8): 3})
assert all(len(cycles) == 2 for cycles in survivors)

```

References

- [1] Jianwei Jiang and Chunhua Yang. Hamilton starters and path decompositions in directed circulants, 2026. arXiv:2609.01256v1.